

Instantaneous Magnetic Field Distribution in Brushless Permanent Magnet dc Motors, Part II: Armature-Reaction Field

Z. Q. Zhu, *Member, IEEE*, and David Howe

Abstract—In Part I an analytical technique for predicting the open-circuit magnetic field distribution in the airgap/magnet region of a brushless permanent magnet dc motor equipped with a surface mounted magnet rotor and a slotless stator was presented. In this paper the analysis is extended to the prediction of the armature reaction field produced by the 3-phase stator windings. As before the motor model is formulated in polar coordinates and accounts for the large effective airgap, but it is now developed further to cater for the effect of winding current harmonics on the airgap field distribution. Predicted instantaneous armature reaction field distributions are validated by comparing with corresponding finite element calculations.

I. INTRODUCTION

A KNOWLEDGE of the armature reaction field calculation in a brushless permanent magnet dc motor is important for the prediction of the radial exciting forces, and hence the emitted acoustic noise [1], the demagnetization withstand capability, and the self- and mutual-winding inductances. However, the calculation differs significantly from that for nonpermanent magnet machines, such as the induction motor, in three aspects. Firstly, due to the low recoil permeability μ_m of permanent magnets—a typical value being $1.1 \mu_0$, machines in which the magnets are mounted adjacent to the airgap present a much larger effective airgap. Secondly, the phase current waveforms contain significant harmonics. Thirdly, the magnetic field produced by the 3-phase stator winding does not rotate uniformly but in incremental steps according to the instants of commutation.

The large effective airgap of a brushless permanent magnet dc motor would lead to a significant error in the armature reaction field calculation if a 1-d analysis in rectangular coordinates, of the form used successfully for induction motors, was used [2]–[5]. Instead a two-dimensional analysis is required to account for flux focusing and leakage effects.

In this paper the analytical field calculation presented in [8], (part I, this issue) is extended to cater for the cal-

culation of the armature reaction field in the airgap/magnet region. The analysis is again in polar coordinates, but now accounts for the effect of the stator winding distribution and current waveform, and the large effective airgap. Predicted field distributions are compared with corresponding finite element calculations.

II. ARMATURE REACTION FIELD

In a similar manner to the prediction of the open-circuit magnetic field described in Part I, the two-dimensional armature reaction field distribution is obtained assuming a smooth airgap in polar coordinates. However a two-dimensional relative permeance function will be introduced in Part III to account for the effect of slotting, thereby permitting the calculation of both the open-circuit and armature reaction field components in slotted radial-field machines having either internal or external rotors. However, for simplicity in the calculation of the armature reaction field the relative recoil permeability of the permanent magnets is assumed to be unity, which will not normally introduce any significant error into the calculation of the load field distribution since the armature reaction field is usually only around 10–20% of the magnitude of the open-circuit field.

A. Magnetic Field Produced by a Coil

Figs. 1(a) and (b) show a distributed current sheet representation, in polar coordinates, of a conductor and a coil respectively, both being accommodated in slots. The current sheet is distributed such that the current density is uniform along an arc whose length is equal to the slot-opening, b_0 .

For simplicity the field of the single conductor, shown in Fig. 1(a) is derived first. The current density distribution is given by

$$J = \begin{cases} \frac{i}{b_0} - \frac{b_0}{2R_s} \leq \alpha \leq \frac{b_0}{2R_s} & \text{(i.e., over the slot-opening)} \\ 0 & \text{other } \alpha \quad \text{(i.e., elsewhere).} \end{cases} \quad (1)$$

Therefore the Fourier series expansion is obtained as

$$J = \frac{i}{2\pi R_s} + \frac{2i}{\pi b_0} \sum_v \frac{1}{v} \sin v \frac{b_0}{2R_s} \cos v\alpha. \quad (2)$$

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The authors are with the Department of Electronic and Electrical Engineering, University of Sheffield, P.O. Box 600, Mappin Street, Sheffield, S1 4DU, United Kingdom.

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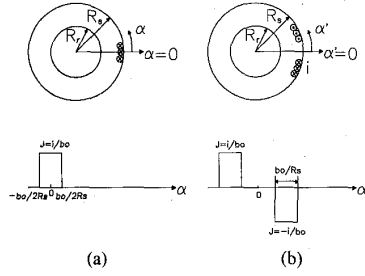


Fig. 1. Distributed current sheet model of conductor/coil in slots. (a) A single conductor. (b) A single coil.

The governing Laplacian equation in polar coordinates is

$$\frac{\partial^2 \varphi}{\partial r^2} + \frac{1}{r} \frac{\partial \varphi}{\partial r} + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \alpha^2} = 0 \quad (3)$$

while if the permeability of the stator core and rotor hub is assumed to be infinite, the boundary conditions to be satisfied are

$$H_\alpha|_{r=R_r} = 0 \quad (4)$$

$$H_\alpha|_{r=R_s} = -J. \quad (5)$$

Since the general solution for $\varphi(\alpha, r)$ is

$$\begin{aligned} \varphi(\alpha, r) = & \sum_v (A_v r^v + B_v r^{-v}) (C_v \cos v\alpha + D_v \sin v\alpha) \\ & + (A_0 \ln r + B_0) (C_0 \alpha + D_0) \end{aligned} \quad (6)$$

and

$$\begin{aligned} H_r &= -\frac{\partial \varphi}{\partial r} & B_r &= -\mu_0 \frac{\partial \varphi}{\partial r} \\ H_\alpha &= -\frac{1}{r} \frac{\partial \varphi}{\partial \alpha} & B_\alpha &= -\mu_0 \frac{1}{r} \frac{\partial \varphi}{\partial \alpha} \end{aligned} \quad (7)$$

the radial component of the magnetic flux density for the distributed current sheet model of the single conductor is obtained as

$$\begin{aligned} B(\alpha, r) &= \mu_0 \frac{i}{\pi} \frac{1}{r} \sum_v \left\{ \frac{\sin v \frac{b_0}{2R_s}}{v \frac{b_0}{2R_s}} \right\} \left(\frac{r}{R_s} \right)^v \frac{1 + \left(\frac{R_r}{r} \right)^{2v}}{1 - \left(\frac{R_r}{R_s} \right)^{2v}} \sin v\alpha \\ &\quad + \frac{i\alpha}{2\pi} \frac{1}{\ln \left(\frac{R_r}{R_s} \right)} \left\{ \ln \left(\frac{R_r}{r} \right) - 1 \right\} \\ &= \frac{\mu_0 i}{\pi} \frac{1}{\delta} \sum_v \frac{1}{v} K_{sov} F_v(r) \sin v\alpha + \frac{i\alpha}{2\pi} \frac{1}{\ln \left(\frac{R_r}{R_s} \right)} \left\{ \ln \left(\frac{R_r}{r} \right) - 1 \right\} \end{aligned} \quad (8)$$

where the reference for the angular variable α , i.e., $\alpha = 0$, is with respect to the center of the slot opening. Similarly, the radial component of the flux density for the distributed current sheet model of the single coil can be de-

rived as

$$B(\alpha, r) = \frac{2\mu_0 i}{\pi} \frac{1}{\delta} \sum_v \frac{1}{v} K_{sov} K_{pv} F_v(r) \cos v\alpha \quad (9)$$

where $\alpha = 0$ corresponds with the axis of the coil, i.e., midway between the two slot openings. Similar expressions can be derived for the circumferential component of flux density in the airgap/magnet region.

In (9) K_{pv} is the winding pitch factor given by $K_{pv} = \sin v \alpha_y / 2$, where α_y is the winding pitch, i.e., coil span, and $\delta = g + h_m$ is the effective airgap length, whilst

$$K_{sov} = \frac{\sin v \frac{b_0}{2R_s}}{v \frac{b_0}{2R_s}} \quad (10)$$

can be appropriately defined as the slot-opening factor. Clearly if the slot-opening b_0 is very small, i.e., $b_0 \rightarrow 0$, then the slot-opening factor approaches unity, i.e., $K_{sov} \rightarrow 1$.

$F_v(r)$ is a function dependant on the radius and harmonic order, and is given by

$$F_v(r) = \delta \frac{v}{r} \left(\frac{r}{R_s} \right)^v \frac{1 + \left(\frac{R_r}{r} \right)^{2v}}{1 - \left(\frac{R_r}{R_s} \right)^{2v}} \quad (11)$$

and will be discussed further in the next section. Equations (8)–(11) apply equally to both internal and external rotor motors. However in the following sections only the analysis of internal rotor motors is considered.

B. Effect of Airgap Length

As was mentioned in the introduction, due to the large effective airgap, the armature reaction field of brushless

permanent magnet dc motors should normally be calculated by a two-dimensional analysis. However it is interesting to compare expressions for the field distribution deduced from both 1-d and 2-d models. If the effective airgap

δ , and hence $|R_s - R_r|$, is small, $r \approx R_r \approx R_s$, therefore

$$\begin{aligned} F_v(r) &= \delta \frac{v}{r} \left(\frac{r}{R_s} \right)^v \frac{1 + \left(\frac{R_r}{r} \right)^{2v}}{1 - \left(\frac{R_r}{R_s} \right)^{2v}} \approx \delta \frac{2v}{R_s} \frac{R_s^{2v}}{R_s^{2v} - R_r^{2v}} \\ &= \delta \frac{2v}{R_s} \frac{R_s^{2v}}{(R_s - R_r)(R_s^{2v-1} + R_s^{2v-2}R_r + \dots + R_r^{2v-1})} \approx \delta \frac{2v}{R_s} \frac{R_s^{2v}}{\delta 2v R_s^{2v-1}} = 1, \end{aligned} \quad (12)$$

i.e., $F_v(r)$ becomes unity, and therefore independent of the radius. Therefore for a small effective airgap and with $b_0 \rightarrow 0$, i.e., $K_{sov} = 1$, and the magnetic field produced by a single coil reduces to:

$$B(\alpha, r) = \mu_0 \frac{2i}{\pi \delta} \sum_{v=1,2,3,\dots}^{\infty} \frac{1}{v} K_{pv} \sin v\alpha \quad (13)$$

which is identical to the result for an induction motor [5]. Therefore, when the effective airgap is sufficiently small, which may be the case when the highest energy product magnets are used, the magnetic field distribution predicted from a 2-d model is identical to that obtained from a simple 1-d model. Furthermore, the armature reaction field produced is identical in both external rotor and internal rotor motors. In both cases, since flux focussing is neglected, it is independent of the radius. However, because the effective airgap in permanent magnet motors is relatively large the armature reaction field should, in general, be predicted by the 2-d analysis, in which the function $F_v(r)$ accounts for the influence of the effective airgap length on the harmonic components of the field produced by a coil or a group of coils. A numerical calculation to illustrate the significance of the effective airgap length on the field distribution is given in Section III.

C. Harmonics in the Phase Winding Current Waveform

In a brushless dc motor the phase winding current waveform contains significant harmonics. A typical waveform, obtained from a dynamic simulation, which may be either analytical or numerical [6], [7], for each of the three phases is shown in Fig. 2. These can be expressed as a Fourier series, viz:

$$\begin{aligned} i_a &= \sum_u I_u \sin [u(p\omega_r t) + \theta_u] \\ i_b &= \sum_u I_u \sin [u(p\omega_r t - 2\pi/3) + \theta_u] \\ i_c &= \sum_u I_u \sin [u(p\omega_r t - 4\pi/3) + \theta_u] \end{aligned} \quad (14)$$

where $\theta = p\omega_r t$, p is the number of pole pairs, ω_r is the rotor angular velocity, and θ_u is the harmonic phase angle.

The waveforms of Fig. 2 together with the harmonic spectrum were obtained from a steady-state dynamic simulation of the 3-phase, 4-pole, radial-field brushless dc motor whose parameters were given in part I [8]. Since the motor has a symmetrical star-connected 3-phase winding and is operating under a steady-state condition, $i(\theta)$

$= -i(\theta + 2\pi)$, and only odd nontriplen order harmonics exist, i.e., $u = 1, 5, 7, 11, 13, \dots$.

D. Armature Reaction Field of 3-Phase Winding

In Section A the magnetic field distribution due to a single current carrying coil accommodated in slots was obtained by a two-dimensional analytical technique. The field of a distributed phase winding can be obtained by introducing a winding distribution factor K_{dpv} , as is commonly applied to 1-dimensional analyses of induction motors [5], i.e.,

$$B(\alpha, r, t) = \mu_0 \frac{2W}{\pi} \frac{i}{\delta} \sum_v \frac{1}{v} K_{sov} K_{dpv} F_v(r) \cos v\alpha \quad (15)$$

where $\alpha = 0$ refers to the axis of the phase winding, W is the number of series turns per phase, and $K_{dpv} = K_{dv} K_{pv}$, where K_{dv} is the winding distribution factor. For example, when the number of slots per pole per phase is an integer, K_{dv} is given by:

$$K_{dv} = \frac{\sin q \frac{v\pi}{Q_s}}{q \sin \frac{v\pi}{Q_s}} \quad (16)$$

where $q = Q_s/2pm$ is the number of stator slots per pole per phase; Q_s is the total number of slots; and m is the number of phases.

Hence the field produced by a 3-phase winding in a brushless dc motor can be deduced from (14) and (15) as:

$$\begin{aligned} B_{winding}(\alpha, r, t) &= B_a(\alpha, r, t) + B_b(\alpha, r, t) \\ &\quad + B_c(\alpha, r, t) \\ &= \mu_0 \frac{2W}{\pi \delta} \sum_v \frac{1}{v} K_{sov} K_{dpv} F_v(r) \\ &\quad \cdot \left[i_a \cos v\alpha + i_b \cos v \left(\alpha - \frac{2\pi}{3} \right) \right. \\ &\quad \left. + i_c \cos v \left(\alpha - \frac{4\pi}{3} \right) \right] \\ &= \mu_0 \frac{W}{\pi \delta} \sum_u I_u \sum_v \frac{1}{v} K_{sov} K_{dpv} F_v(r) \end{aligned}$$

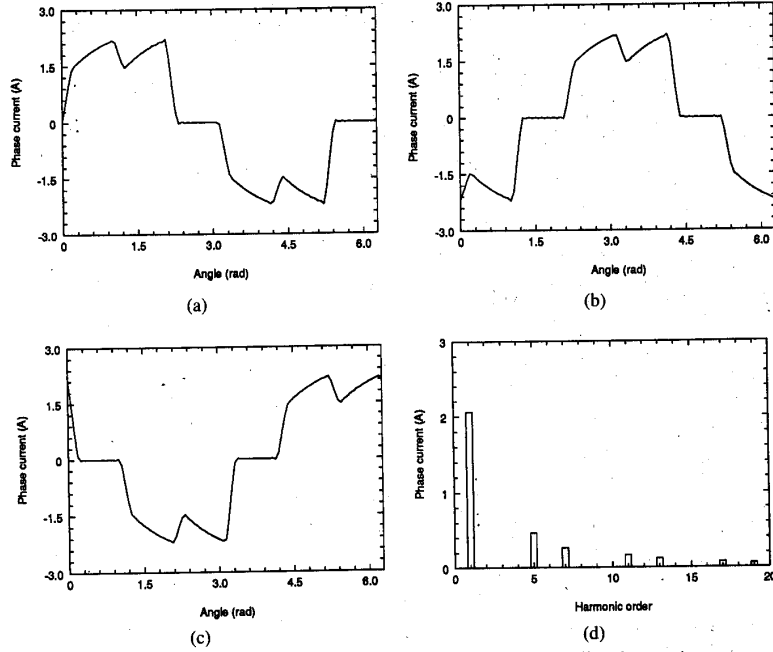


Fig. 2. Phase winding current waveforms and corresponding harmonic spectrum for 3-phase, 4-pole brushless dc motor running at 1500 rpm. (a) Phase a. (b) Phase b. (c) Phase c. (d) Harmonic spectrum.

$$\begin{aligned}
 & \cdot 0.2 \left\{ \sin [u(p\omega_r t) + \theta_u] \cos v\alpha \right. \\
 & + \sin [u(p\omega_r t - 2\pi/3) + \theta_u] \\
 & \cdot \cos v \left(\alpha - \frac{2\pi}{3p} \right) \\
 & + \sin [u(p\omega_r t - 4\pi/3) + \theta_u] \\
 & \cdot \cos v \left(\alpha - \frac{4\pi}{3p} \right) \left. \right\} \quad (17)
 \end{aligned}$$

where

$$\begin{aligned}
 2\{\} &= \begin{cases} \sin [(up\omega_r t) + \theta_u + v\alpha] \\ + \sin \left[(up\omega_r t) + \theta_u + v\alpha - \left(\frac{v}{p} + u \right) 2\pi/3 \right] \\ + \sin \left[(up\omega_r t) + \theta_u + v\alpha - \left(\frac{v}{p} + u \right) 4\pi/3 \right] \end{cases} \\
 &+ \begin{cases} \sin [(up\omega_r t) + \theta_u - v\alpha] \\ + \sin \left[(up\omega_r t) + \theta_u - v\alpha + \left(\frac{v}{p} - u \right) 2\pi/3 \right] \\ + \sin \left[(up\omega_r t) + \theta_u - v\alpha + \left(\frac{v}{p} - u \right) 4\pi/3 \right] \end{cases} \\
 &= \left[1 + 2 \cos \left(\frac{v}{p} + u \right) 2\pi/3 \right] \\
 &\cdot \sin \left[(up\omega_r t) + \theta_u + v\alpha - \left(\frac{v}{p} + u \right) 2\pi/3 \right]
 \end{aligned}$$

while

$$\begin{aligned}
 & \frac{\sin \left(\frac{v}{p} \pm u \right) \pi}{\sin \left(\frac{v}{p} \pm u \right) \pi/3} = \begin{cases} 0 & \text{for } \frac{v}{p} \pm u \neq 3c \\ c & \text{for } \frac{v}{p} \pm u = 3c \\ 3 & \text{for } \frac{v}{p} \pm u = 3c \end{cases} \quad (19)
 \end{aligned}$$

$$\begin{aligned}
 & + \left[1 + 2 \cos \left(\frac{v}{p} - u \right) 2\pi/3 \right] \\
 & \cdot \sin \left[(up\omega_r t) + \theta_u - v\alpha + \left(\frac{v}{p} - u \right) 2\pi/3 \right] \\
 & \frac{\sin \left(\frac{v}{p} + u \right) \pi}{\sin \left(\frac{v}{p} + u \right) \pi/3} \\
 & \cdot \sin \left[(up\omega_r t) + \theta_u + v\alpha - \left(\frac{v}{p} + u \right) 2\pi/3 \right] \\
 & + \frac{\sin \left(\frac{v}{p} - u \right) \pi}{\sin \left(\frac{v}{p} - u \right) \pi/3} \\
 & \cdot \sin \left[(up\omega_r t) + \theta_u - v\alpha + \left(\frac{v}{p} - u \right) 2\pi/3 \right] \quad (18)
 \end{aligned}$$

Therefore

$$\begin{aligned} 2\{\} &= 3 \sin [(up\omega_r t + \theta_u \pm v\alpha) - \{\pm\} 3c(2\pi/3)] \\ &= 3 \sin (up\omega_r t + \theta_u \pm v\alpha) \end{aligned} \quad (20)$$

and

$$\frac{v}{p} \pm u = 3c,$$

i.e.,

$$v = p(3c - \{\pm\}u). \quad (21a)$$

However, a symmetrical 3-phase winding with equal phase-belts produces only harmonics of the order

$$v = p(6c - \{\pm\}u) \quad (21b)$$

Hence the two-dimensional analysis of the magnetic field is obtained by combining (17), (20), and (21) to give:

$$\begin{aligned} B_{\text{winding}}(\alpha, r, t) &= \mu_0 \frac{3W}{\pi} \frac{1}{\delta} \sum_u I_u \sum_v \frac{1}{v} K_{sov} K_{dpv} F_v(r) \\ &\quad \cdot \sin [up\omega_r t \pm v\alpha + \theta_u] \end{aligned} \quad (22)$$

where $\alpha = 0$ corresponds to the axis of the phase A winding, whose current is zero at $t = 0$, i.e., $i_a|_{t=0} = 0$, but which is commutated on at $t = 0$.

It is worth noting that although the above approach is being applied specifically to 3-phase brushless permanent magnet dc motors, it is equally applicable to the analysis of the magnetic field produced by the stator windings of any variable speed ac motor, simply by customizing it to model the particular motor format. For example, the analysis is directly applicable to brushless ac permanent magnet motors, whilst by setting δ equal to the actual airgap length and $F_v(r) = 1$ it can be applied to an inverter controlled 3-phase induction motor.

III. COMPARISON WITH FINITE ELEMENT CALCULATIONS

Stator slotting affects both the magnitude and distribution of the armature reaction field. However, as in part I [8] the effects are neglected for the time being, until the modelling technique described in part III [9] is developed.

Fig. 3 shows the finite element mesh for the 4-pole internal rotor motor whose parameters are given in part I [8] and whose current waveforms are shown in Fig. 2, together with flux distributions due to the winding excitation alone for three different instantaneous rotor positions. As with the analytical model, stator slotting has been neglected and the tooth tips have been modelled by a natural Neuman boundary condition, the instantaneous winding currents being represented as equivalent current sheets distributed over an arc of the stator bore corresponding to the slot openings. It can be seen that over a large portion of the airgap/magnet region the flux paths deviate considerably from the radial direction, which again emphasizes the need for a two-dimensional analysis.

Fig. 4 compares analytically derived flux density dis-

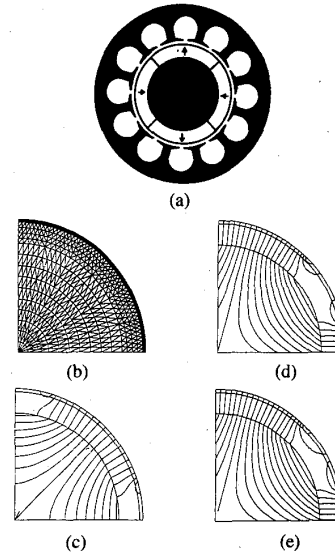


Fig. 3. Finite element mesh and instantaneous flux distributions due to the winding excitation alone. (a) Slotted motor cross-section. (b) Finite element mesh of equivalent slotless motor. (c) Flux distribution at time = 0 s. (d) Flux distribution at time = 1.667 ms. (e) Flux distribution at time = 3.333 ms.

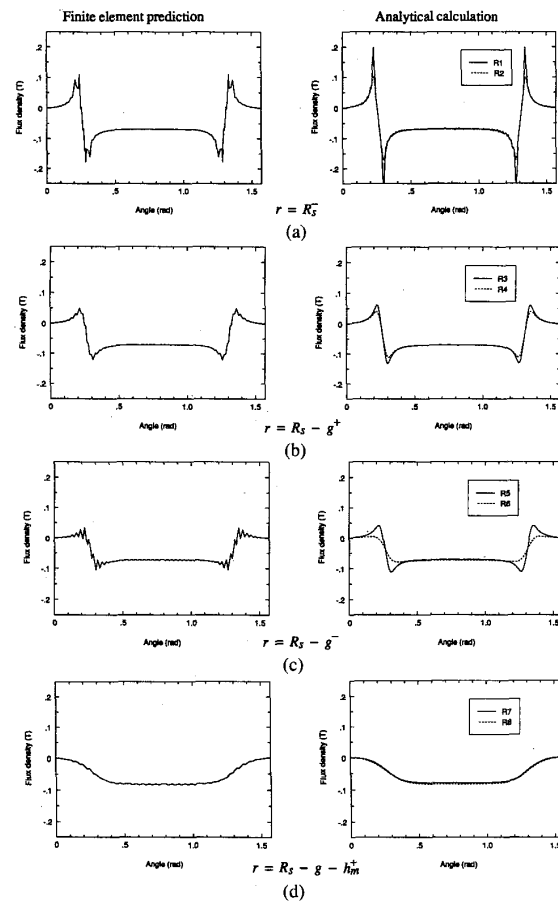


Fig. 4a. Comparison of field distribution produced by the windings at different radii, time = 0 s. (a) $r = R_s^-$. (b) $r = R_s - g^+$. (c) $r = R_s - g^-$. (d) $r = R_s - g - h_m^+$. The definition of radii is given in Table I of Part I [8].

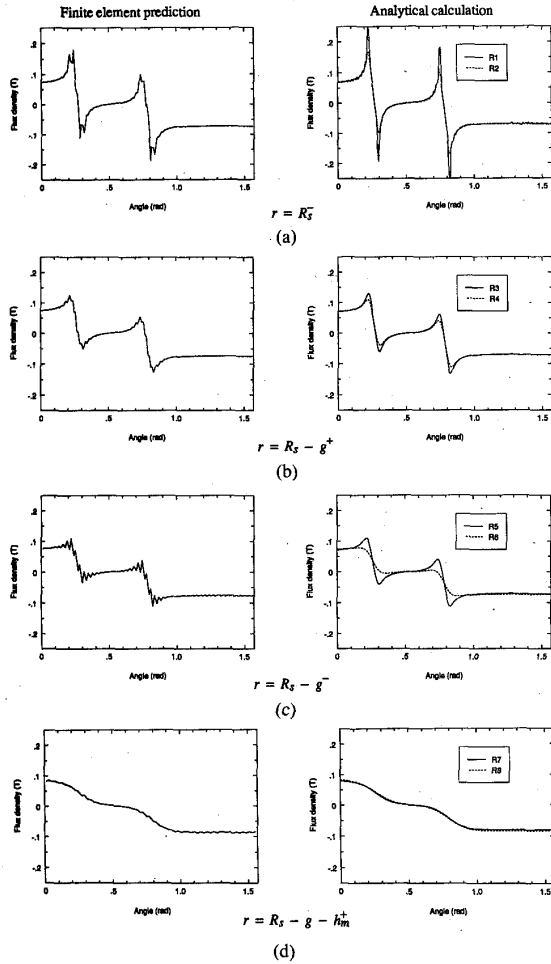


Fig. 4b. Comparison of field distribution produced by the windings at different radii, time = 3.333 ms. (a) $r = R_s^-$. (b) $r = R_s - g^+$. (c) $r = R_s - g^-$. (d) $r = R_s - g - h_m^+$. The definition of radii is given in Table I of Part I [8].

tributions at various radii in the airgap/magnet region with corresponding finite element calculations. As in part I [8], analytical predictions are included for the two radii which correspond to the nodes of each layer of finite elements, the radii being given in Table I of part I [8]. The analytical predictions and the finite element calculations show excellent agreement.

However, it is also interesting to compare the above results with those calculated by a 1-d analytical method and shown in Fig. 5. The main differences are that:

- 1) The flux density distribution now has a stepped rectangular waveform, the incremental variation in airgap flux density in moving past a current carrying slot being:

$$\Delta B = \frac{\mu_0 N_s i}{\delta} \quad (23)$$

for full pole-pitch windings, where N_s is the number of conductors in the slot, δ is the effective airgap

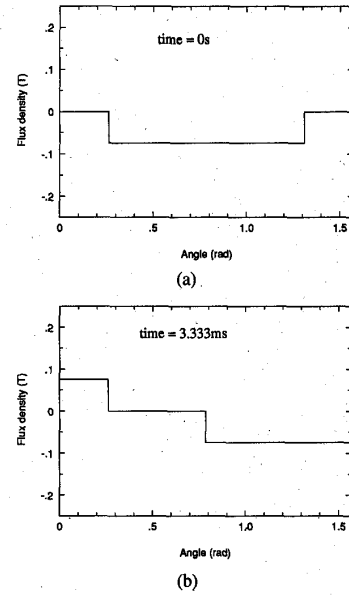


Fig. 5. Instantaneous field distribution produced by the windings from a 1-d analytical prediction. (a) time = 0 s. (b) time = 3.33 ms.

length, and i is the instantaneous current. For example, for this particular internal rotor motor having $N_s = 150$ and $\delta = 5.45$ mm, at time $t = 0$ when $i = 2.15$ A, $\Delta B = 0.075$ T.

- 2) The flux density distribution is independent of the radius. In contrast, a 2-dimensional analysis shows that a) The flux density distribution changes in passing from the rotor hub to the stator bore, and there is significant flux focussing at the tips of the stator teeth adjacent to the winding current sheets. b) The amplitude of the flux density varies with the radius due to the combined effects of flux focussing and flux leakage. For the particular internal rotor motor which has been analyzed the flux density is lowest near the axis of the stator coils and greatest at the surface of the rotor hub. However, for an external rotor radial-field motor topology the flux density will always decrease in moving from the stator to the rotor hub.

It will be noted that, as in part I [8], the oscillations in the flux density predictions result from discretization effects in the finite element analyses and from the use of a finite number of terms in the Fourier series of the analytical calculations.

IV. CONCLUSIONS

A 2-d analytical technique modelled in polar coordinates has been developed for predicting the magnetic field produced by the stator windings in a radial-field 3-phase brushless dc motor, accounting implicitly for the stator winding current waveform. The predicted field distributions show excellent agreement with results from finite

element analyses. It has also been shown that, due to their much larger effective airgap, a traditional 1-d model of the type used for more conventional machine formats is unsuitable.

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Z. Q. Zhu (M'90) was born in Zhejiang, China, in 1962. He received the B.Eng. and M.Sc. degrees from Zhejiang University, China, in 1982 and 1984, respectively, and was awarded the Ph.D. from the University of Sheffield in 1991, all in electrical and electronic engineering.

From 1984 to 1988 he worked in the Department of Electrical Engineering at Zhejiang University, being appointed a lecturer in 1986. Since 1988 he has been a research associate in the Department of Electronic and Electrical Engineering at the University of Sheffield, working on CAD, electromagnetic field computation, and acoustic noise prediction, particularly in relation to permanent magnet machines, actuators, and drives.

David Howe was born in Sheffield, England in 1943. He received the B.Tech. and M.Sc. degrees from the University of Bradford in 1964 and 1967, respectively, and the Ph.D. from Southampton University in 1974.

He has held academic posts at Brunel and Southampton Universities, and spent a period in industry with NEI Parsons Ltd. working on electromagnetic problems in turbo-generators. He is currently Professor of Electrical Engineering at the University of Sheffield where he heads the Electrical Machines & Drives Research Group. His research activities span all facets of controlled electrical drive systems, with particular emphasis on permanent magnet excited machines.

Professor Howe has published extensively.